



CBCS SCHEME

BESCK104C/BESCKC104

First Semester B.E/B.Tech. Degree Examination, June/July 2024 Introduction to Electronics and Communication

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
1	a.	Explain the working of Bi-phase full wave rectifier circuit with a neat diagram and waveform.	7	L2	CO1
	b.	A zener diode has a breakdown voltage of 10V. It is supplied from a voltage source varying between 20 – 40V in series with a resistance of 820Ω . Using an ideal zener model obtain the minimum and maximum zener currents.	7	L2	CO1
	c.	Summarize the advantages of negative feedback in amplifier circuits. With relevant equations and diagram, explain the concept of negative feedback.	6	L2	CO1
OR					
2	a.	Describe the need of capacitor filter. For a FWR, explain the operation of C-filter.	7	L2	CO1
	b.	Discuss the load and line regulation using zener diode with neat circuit diagram and appropriate expressions.	7	L2	CO1
	c.	Describe with relevant frequency response curve of RC coupled amplifier. Explain the reason for the fall of voltage gain at low and high frequencies.	6	L2	CO1
Module – 2					
3	a.	What is Barkhausen criteria for sustained oscillations? How it is used in oscillator?	7	L1	CO2
	b.	Explain with the help of circuit diagram the operation of a crystal oscillator. Why these oscillators give highly stable oscillations. Mention applications of crystal oscillators.	7	L2	CO2
	c.	Sketch the circuit of each of the following and briefly explain them based on the use of operational amplifiers. i) Differentiator ii) Integrator.	6	L2	CO2
OR					
4	a.	Explain the operation of single stage Astable oscillator with its circuit diagram.	7	L2	CO2
	b.	A sinusoidal signal with peak value 6mv and 2KHz frequency is applied to the i/p of an ideal op-amp integrator with $R_1 = 100K\Omega$ and $C_f = 1\mu f$. Find the o/p voltage.	7	L2	CO2
	c.	List the various ideal op-amp characteristics.	6	L1	CO2

Module – 3

5	a.	Covert : i) $(725.25)_8$ to its decimal and binary equivalent ii) Determine the value of x if $(211)_x = (152)_8$ iii) Realize an OR logic gate using diodes.	7	L2	CO3
	b.	Illustrate how NAND gate can be used to realize the following gates : i) NOR ii) EX – OR.	7	L2	CO3
	c.	Simplify and realize the following expressions using only NAND and NOR. i) $Y = (A + \bar{B})(B + C)(\bar{C} + B)$ ii) $Y = AB + AC + BD + CD$.	6	L2	CO3

OR

6	a.	Prove NAND and NOR is not associative.	6	L2	CO3
	b.	Enumerate the ruler of Boolean algebra and prove each of them with truth table.	7	L2	CO3
	c.	Explain full adder circuit with truth table. Realize the circuit for sum and carry using basic gates. Also write diagram showing FA using two HA.	7	L2	CO3

Module – 4

7	a.	Outline on transducers, sensors and actuators with examples for each.	7	L1	CO4
	b.	Explain the classification of embedded system based on generation.	7	L1	CO4
	c.	Identify the difference between microprocessor and microcontroller.	6	L1	CO4

OR

8	a.	Explain instrumentation system with relevant diagram.	7	L2	CO4
	b.	Explain the working operation of LED with a suitable diagram.	7	L2	CO4
	c.	Construct the block diagram for control system and explain it.	6	L2	CO4

Module – 5

9	a.	Outline on different types of modulation and briefly describe each in detail.	7	L1	CO5
	b.	Explain the concept of radio wave propagation and briefly describe each in detail.	7	L1	CO5
	c.	Construct ASK, FSK and PSK waveform by considering the following binary data : (Refer Fig.Q9(c)).	6	L2	CO5



Fig.Q9(c)

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OR

10	a.	Define frequency deviation and sketch the FM wave with illustration.	7	L2	CO5
	b.	Classify the advantages of analog over digital communication.	7	L1	CO5
	c.	Model the architecture of a wireless communication transmitter and its modulators scheme QPSK with waveform and constellation diagram.	6	L2	CO5
